### The VLSI layout

From a fabrication standpoint though, the most significant view is the standard cell's VLSI layout. This is because it's the closest that things get to a manufacturing blueprint for the cell.

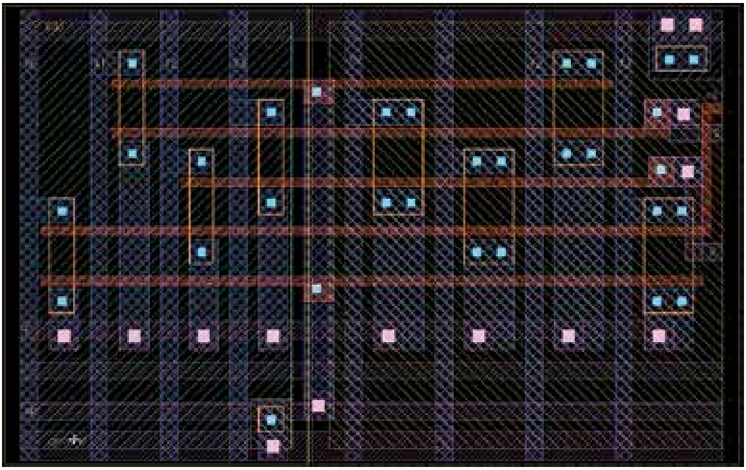
The layout typically has base layers that correspond to the various structures of the transistor devices. It will also have the interconnect wiring layer as well as 'via layers'- they club together the multiple terminals of the transistor formations. Generally, the interconnect wiring layers are numbered. They would also give precise via layers that represent connections between specific sequential layers.

In this layout, it's possible that non-manufacturing layer may be present as well. This is for the purpose of design automation. However, multiple layers that are used specifically for Place and Route(PNR) CAD programs too generally get included in a different yet somewhat similar abstract view.

More often than not, the abstract view has significantly lesser volume of information compared to the layout. It may also be recognized as a Layout Extraction Format (LEF) or equivalent file type.

## The Library

A standard cell library refers to a collection of low-level logic functions like OR, AND, latches, flip-flops, INVERT and buffers. These cells get realized as fixed-height but variable-width completely custom cells. With these



A layout to put things in perspective